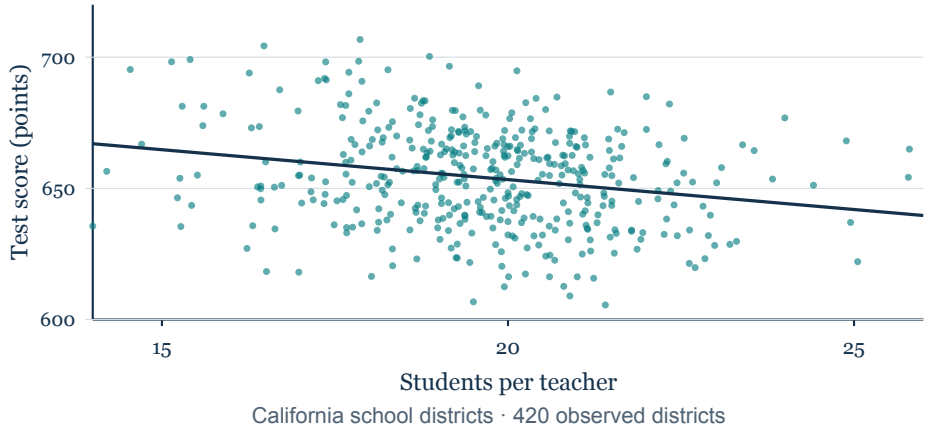
# Do smaller classes improve learning?

Compare districts. Then ask what else differs.




California school districts · 420 observed districts

**Practice**

Do you expect the line to slope upward or downward? Name one other district characteristic that could help explain the pattern.

**Explanation:** A downward slope is plausible: districts with more students per teacher may have lower scores. Family resources could also matter. For example, districts serving richer families may have both more teachers and higher scores. The graph lets us check the observed pattern; it does not yet isolate the effect of hiring teachers.

# Which question are we answering?

The same data can describe a pattern, predict an outcome or help study a policy effect.

> Description: how do observed districts differ?

> Prediction: how well can we forecast a new outcome?

> Causation: what changes when the policy changes?

# Define the observation before the equation

One observation is one district; define the variables and their units.

> $i$: school district

> $Y_i$: average score, in points

> $X_i$: students per teacher

> $n = 420$ districts

**Practice**

Would ten years of the same 420 districts form a panel? Would ten different districts observed each year necessarily do so?

**Explanation:** The first is a panel if district identities are tracked. The second may be repeated cross sections; the calendar alone does not create within-unit observations.

# Start with a rule for how the outcome is produced

A causal model states mechanisms before asking what would change.

> Staffing: $X = 22 - C + V$

> Learning: $Y = 700 - 2X + 5C + U$

> $C = 2, V = 2, U = 3 \Rightarrow X = 22, Y = 669$

# Define the intervention before its counterfactual

Change the staffing rule; retain the stated learning mechanism.

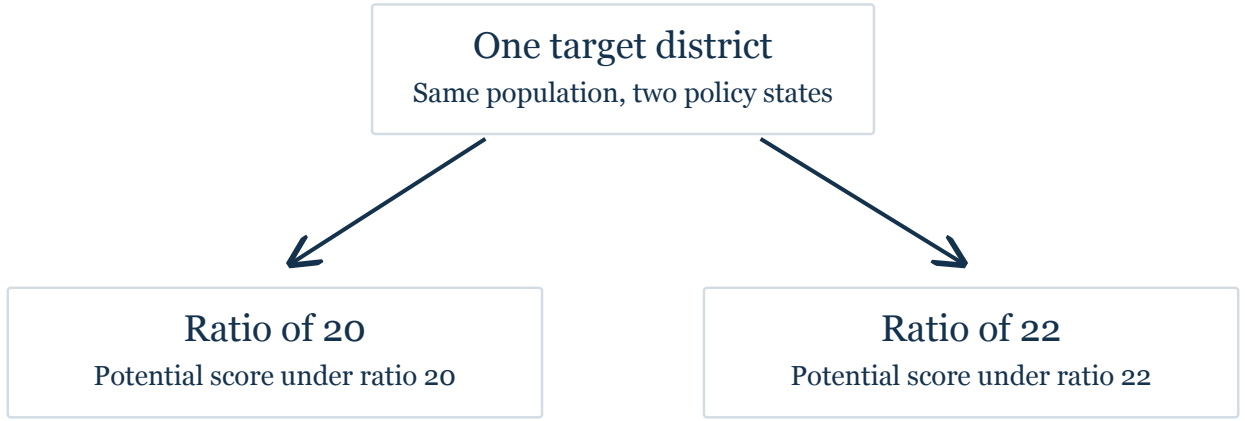
> Intervene: replace the assignment rule by $X = 20$

> Alternative score: $700 - 2(20) + 5(2) + 3 = 673$

> Same-district difference: $673 - 669 = 4$ points

# The comparison we want is partly missing

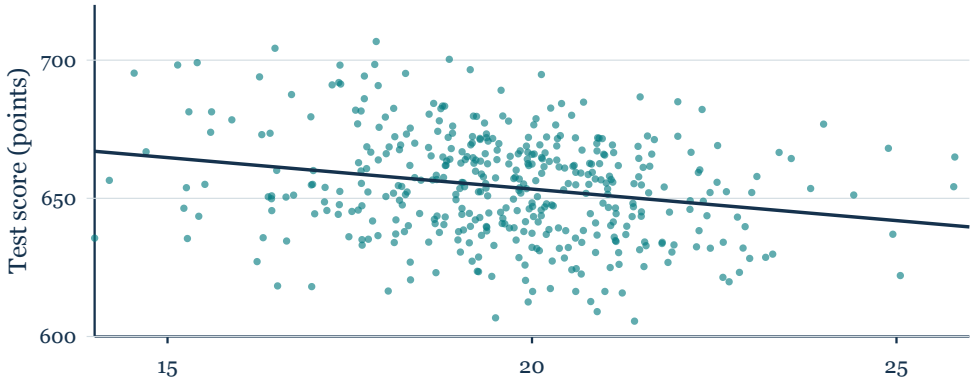
A policy effect compares potential outcomes under two staffing choices.




- Potential outcomes: $Y_i(20), Y_i(22)$
- Causal contrast: $Y_i(20) - Y_i(22)$
- Observed groups do not automatically supply the missing outcome.

# Read the data before fitting a line

Start with axes, units, range and unusual observations.




California school districts · 420 observed districts

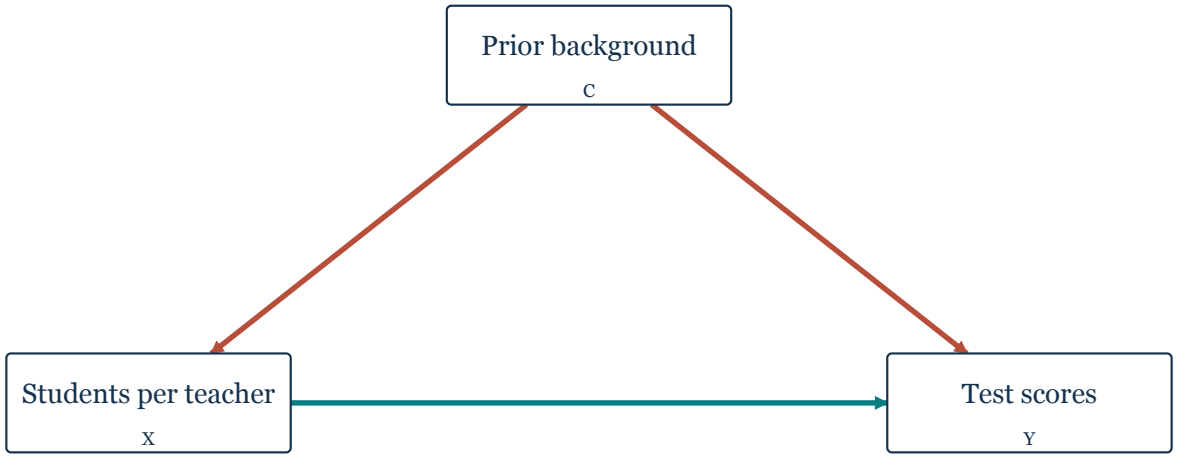
**Practice**

Districts at about the same student–teacher ratio have different scores. Does this prevent a line from describing their average pattern?

**Explanation:** No. A line can summarise the average pattern even when individual districts lie above and below it. Those differences tell us that staffing alone does not fully explain scores.

# Use a DAG to make the causal story visible

Begin with assumptions about arrows before choosing controls.



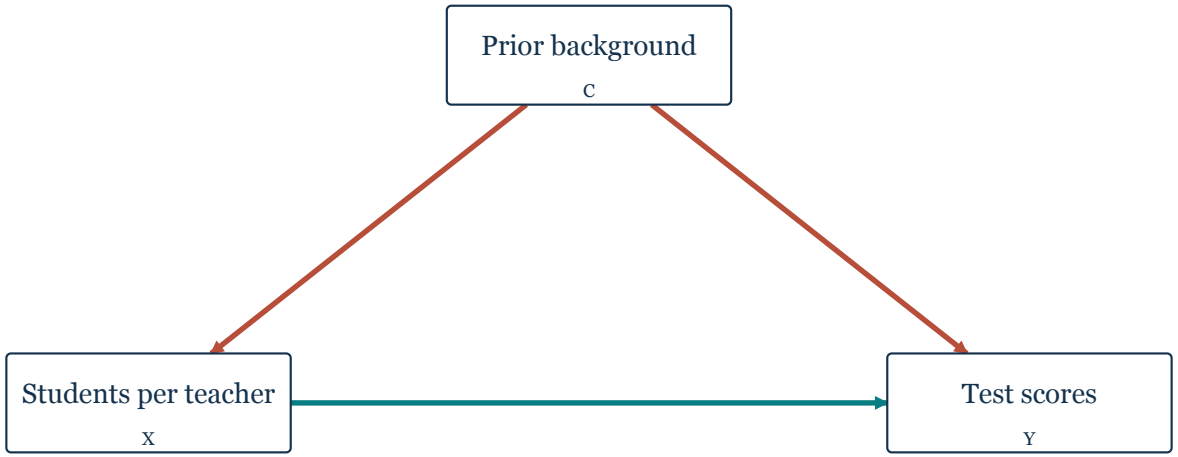

1 open directed route · 1 open alternative · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

- Node: a variable; arrow: a proposed direct causal mechanism.
- Directed: arrows have a direction. Acyclic: no directed loop.
- Missing arrows and independent background inputs are assumptions.

# Trace a path one connection at a time

A causal route follows arrows; a general path need not.



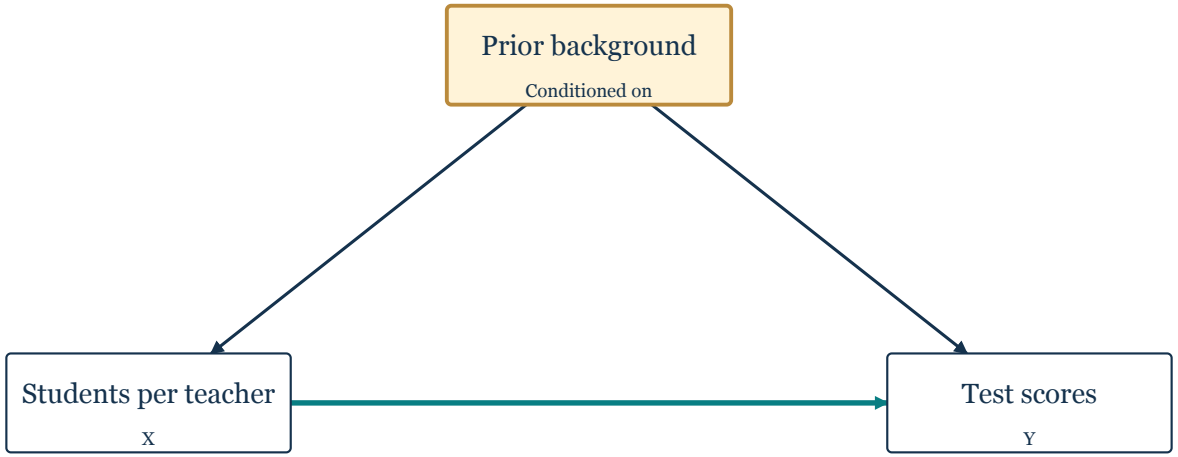

1 open directed route · 1 open alternative · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

- Causal route: $X \to Y$
- Backdoor route: $X \leftarrow C \to Y$

# A common cause creates a backdoor comparison

Compare districts with similar prior background.



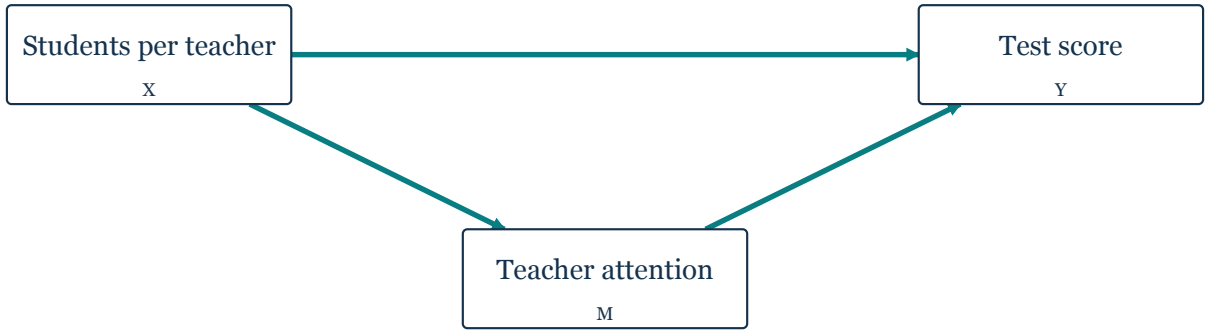

1 open directed route · 0 open alternatives · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

- Compare at fixed $C$: block $X \leftarrow C \rightarrow Y$
- Conditioning selects comparisons; intervention changes a mechanism.

# An intermediate mechanism is a mediator

A variable caused by staffing can carry part of its effect.




2 open directed routes · 0 open alternatives · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

- Mediated route: $X \to M \to Y$
- Total effect allows the mediator to respond.

# Selection can create a relationship

A four-case scholarship example makes the mechanism visible.

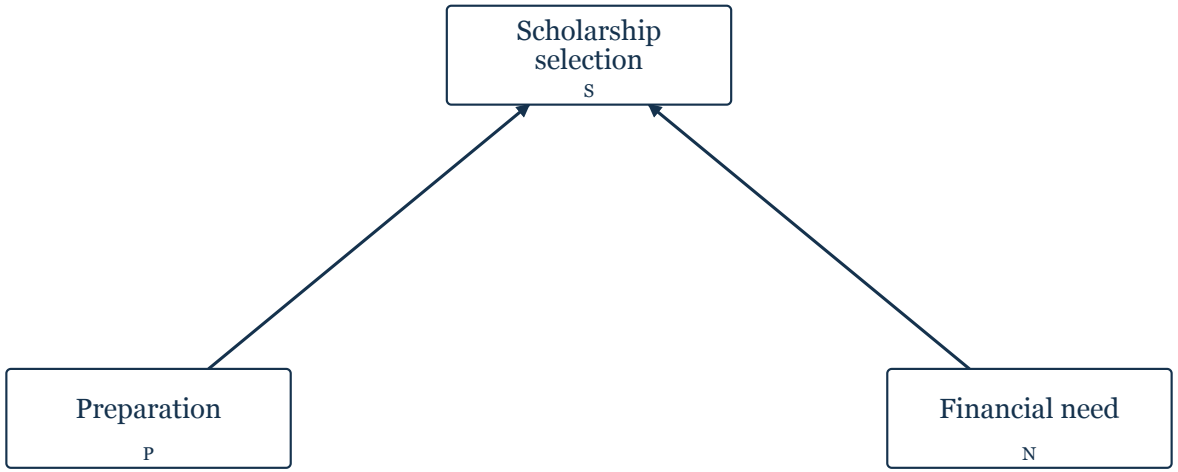
> Equally likely $(P, N)$: $(0,0), (0,1), (1,0), (1,1)$

> Award if $P = 1$ or $N = 1$; exclude only $(0,0)$

> Among recipients: $\Pr(N = 1 \mid P = 0) = 1$, $\Pr(N = 1 \mid P = 1) = 1/2$

# A shared consequence behaves differently

Conditioning can open a route that was blocked.




0 open directed routes · 0 open alternatives · gold: conditioned on

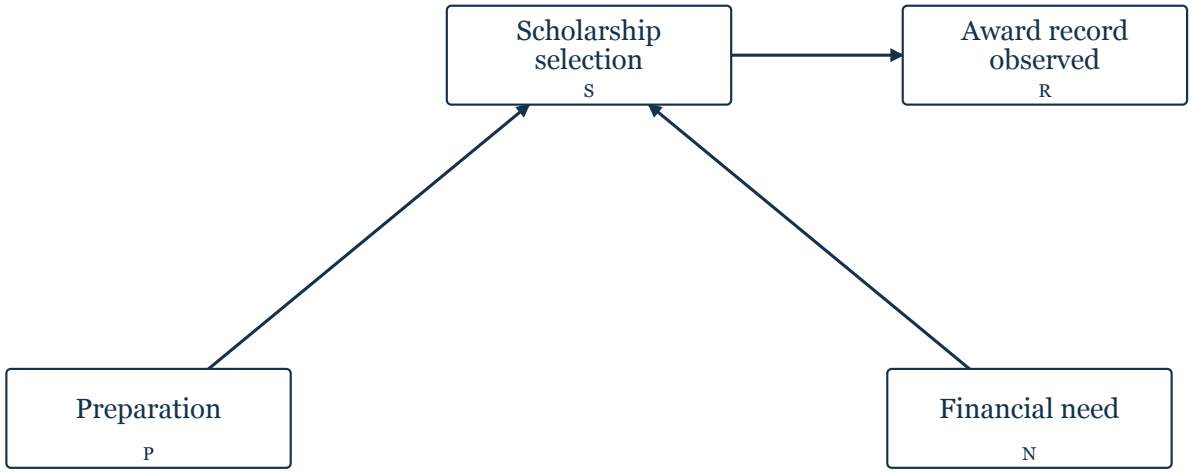
Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

> Collider: $P \rightarrow S \leftarrow N$

> Unconditioned collider blocks; selecting recipients opens the path.

# Selecting a consequence can also open the route

A descendant is reached by following arrows forward.




0 open directed routes · 0 open alternatives · gold: conditioned on

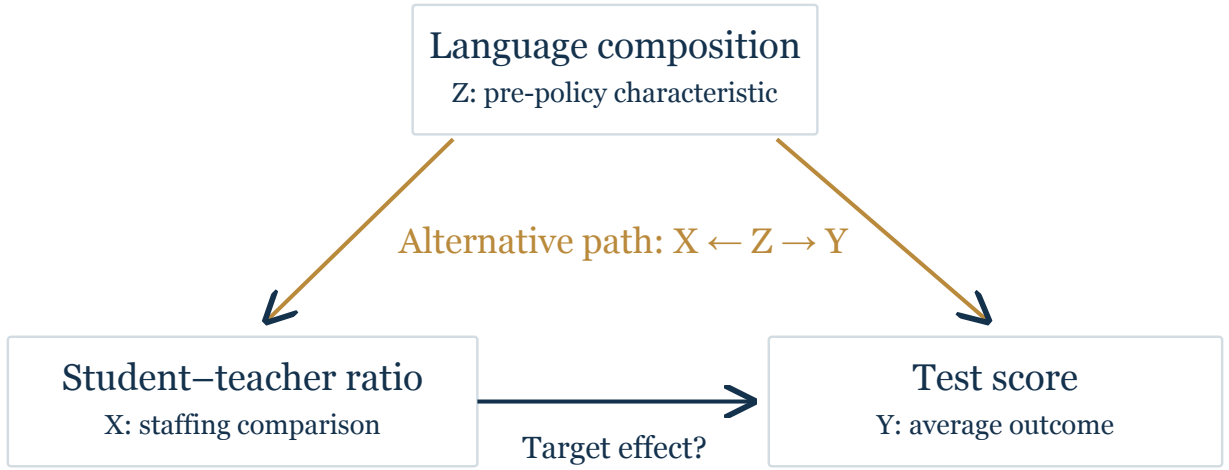
Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

> $P \to S \leftarrow N$, with $S \to R$

> Conditioning on a collider descendant can open its path.

# Why the observed groups might differ

A confounder is another factor related to both staffing and scores.



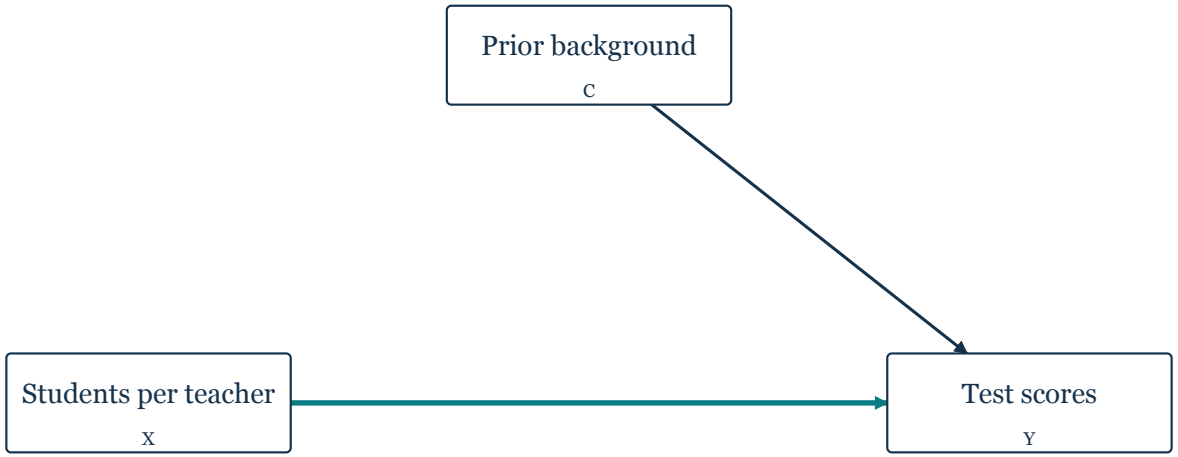

Proposed causal structure, not a relationship proved by the scatterplot.

Arrows show the proposed explanation. The scatterplot alone does not establish it.

- $Z$: English-learner share, in percent
- Possible confounder: related to staffing and scores
- $X \leftarrow Z \rightarrow Y$: an alternative explanation

# An intervention graph changes a different object

The policy removes incoming staffing arrows rather than selecting a subgroup.




1 open directed route · 0 open alternatives · gold: conditioned on

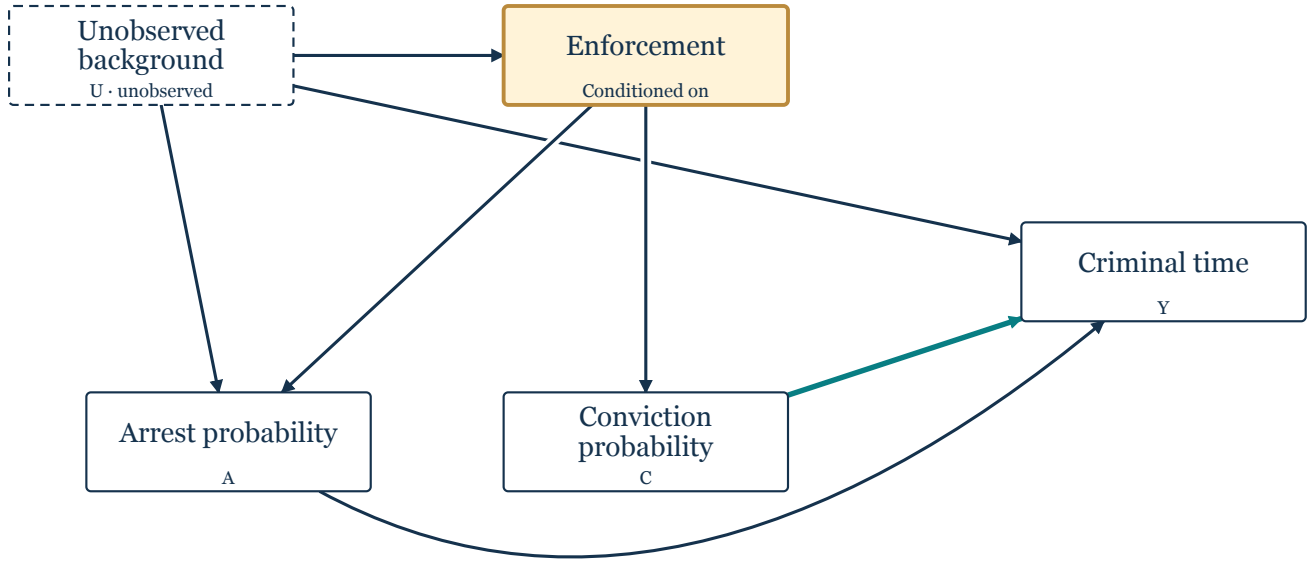
Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

> Intervention: replace the assignment mechanism.

> Conditioning: compare within an observed subgroup.

# Law and order: identify the conviction comparison

A useful adjustment closes every drawn backdoor route.




1 open directed route · 0 open alternatives · gold: conditioned on

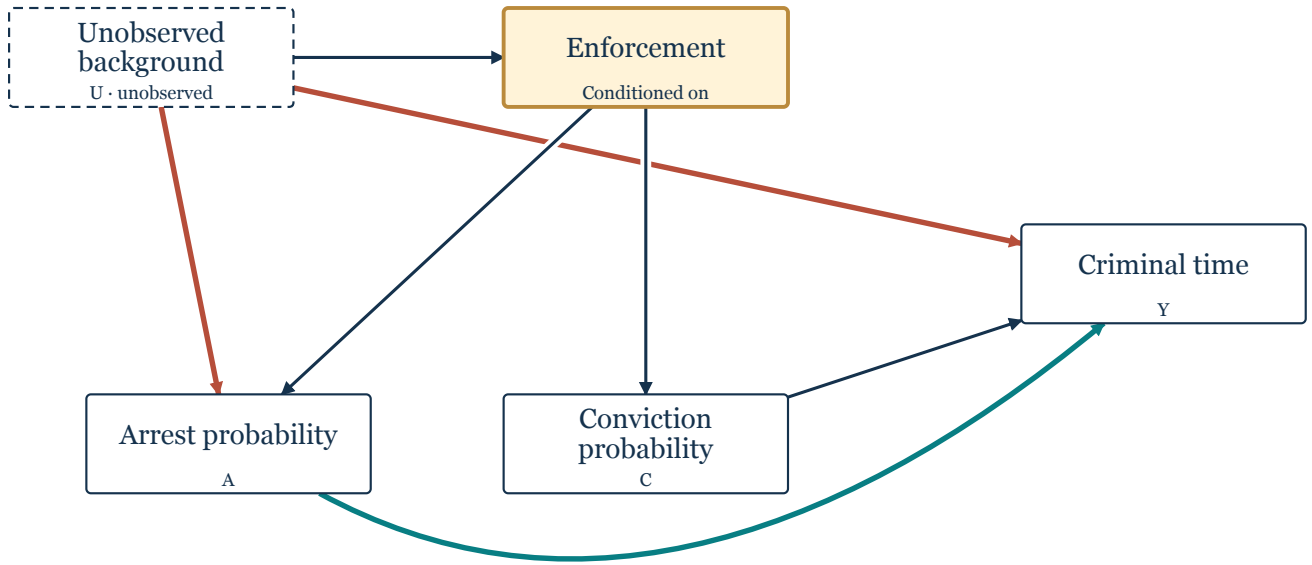
Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

> Target: conviction probability → criminal time

> Ask which variable closes every backdoor route.

# Changing the target can change the required controls

The arrest comparison still has an unobserved route.




1 open directed route · 1 open alternative · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

Remaining route: $A \leftarrow U \rightarrow Y$

A control plan is specific to the chosen explanatory variable.

# Make the first regression model explicit

Define the regression equation, its coefficients and the error term.

> Population: $Y_i = \beta_0 + \beta_1 X_i + u_i$

> Conditional-mean assumption: $E(u_i \mid X_i) = 0$

> Fit: $\widehat{Y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 X_i$; residual: $e_i = Y_i - \widehat{Y}_i$

# A slope is a change with units

A fitted change equals the slope times the change in the regressor.

> $\Delta X = \text{new ratio} - \text{old ratio}$

> $22 \to 20$ gives $\Delta X = -2$

> Fitted change: $b_1 \Delta X \approx +4.56$ score points

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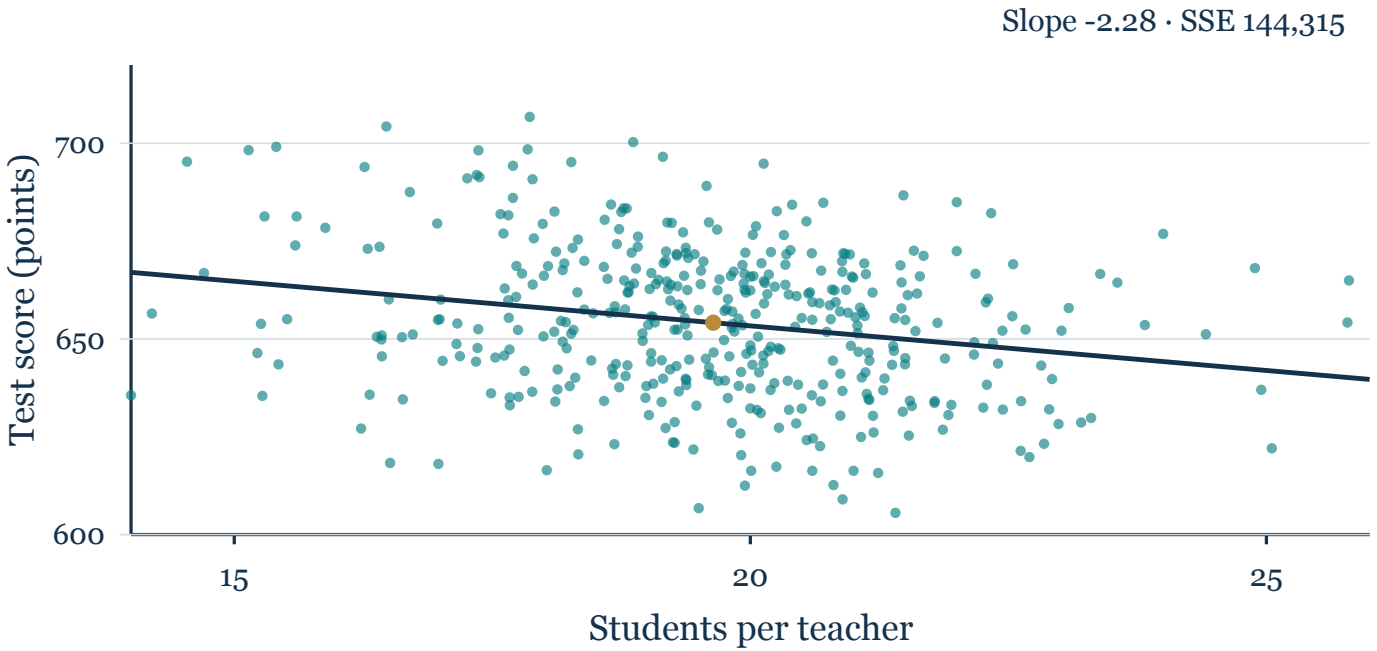
**Practice**

Use the same slope to calculate the fitted score change when the ratio rises from 20 to 23.

**Explanation:** The change in the ratio is +3. The fitted score change is 3×(−2.279808), about −6.84 score points. Moving right along the downward-sloping line lowers its fitted score.

# Choose a line by its errors

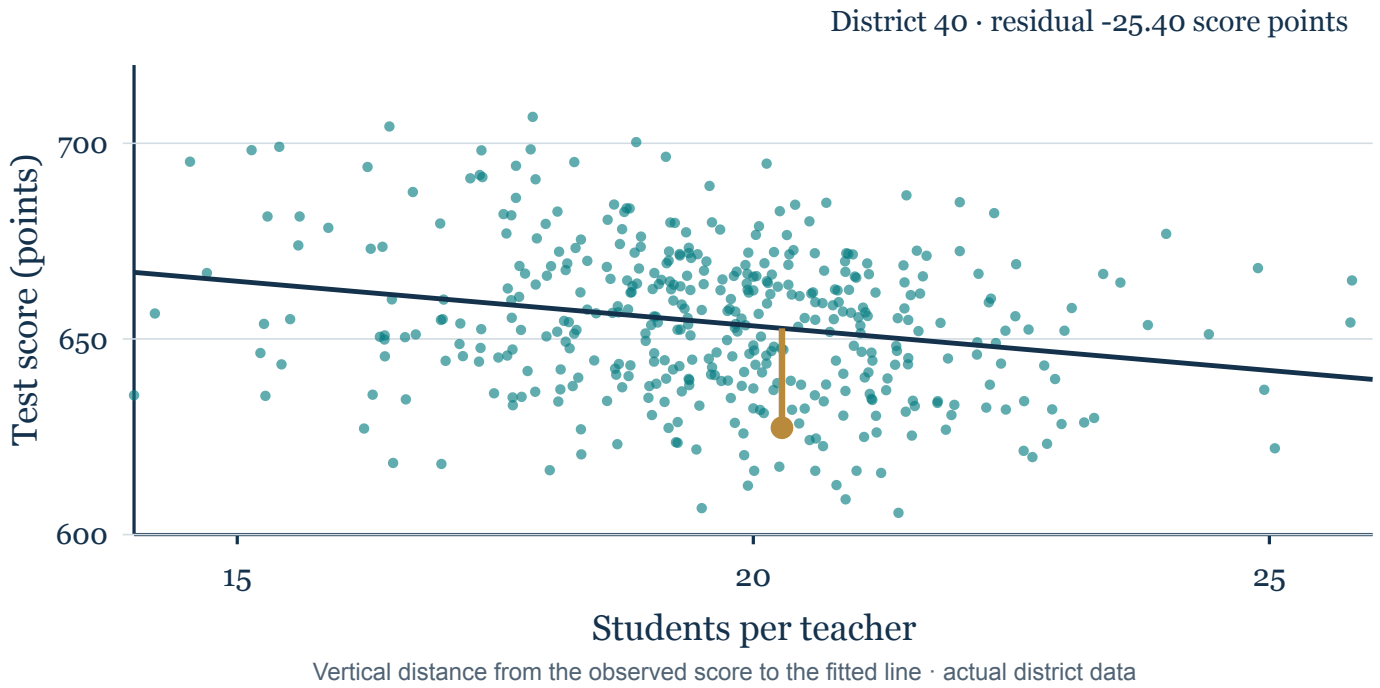
OLS chooses the line with the smallest sum of squared residuals.




Rotate around the mean point. The loss uses all 420 observed districts.

# Follow one residual

For one district, compare the observed score with the score on the line.



# Calculate the OLS slope from the data

Compare each variable with its average, then combine the deviations.

$$b_1 = \frac{\sum_i (X_i - \overline{X})(Y_i - \overline{Y})}{\sum_i (X_i - \overline{X})^2}$$

$$b_0 = \overline{Y} - b_1 \overline{X}$$

Without regressor variation, the slope cannot be separately estimated.

# A small calculation you can finish

Use three constructed districts to see the formula work.

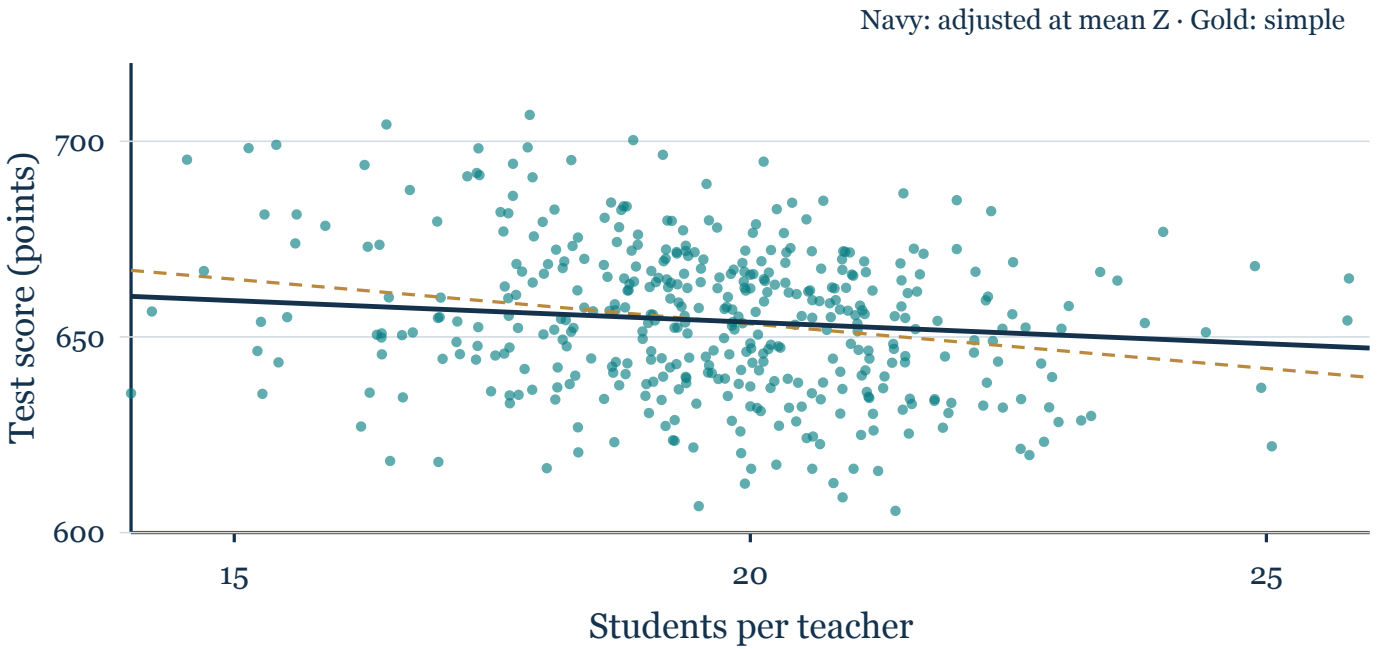
**Practice**

Find the slope, intercept, fitted scores and residuals for X=(18,20,22) and Y=(680,670,650).

**Explanation:** $b_1$=−7.5 and $b_0$=816⅔. Fitted scores are 681⅔, 666⅔ and 651⅔. Residuals are −1⅔, +3⅓ and −1⅔, which sum to zero. Small differences from rounding are acceptable.

# Holding an observed factor fixed

Multiple regression compares ratios and scores at a fixed value of another variable.

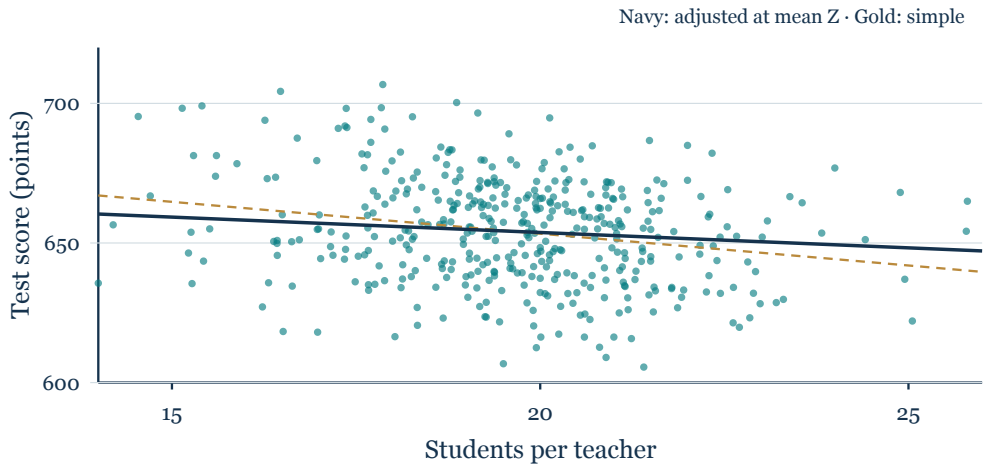



Adjusted line fixes English-learner share at its sample mean. Both fits use the same 420 districts.

- $\widehat{Y_i} = b_0 + b_1 X_i + b_2 Z_i$
- $b_1$: staffing slope at fixed $Z$
- The graph fixes language share at its sample average.

# Explain why the estimated slope changed

Adding Z changes the fitted comparison from about +4.56 to +2.20 score points.




Adjusted line fixes English-learner share at its sample mean. Both fits use the same 420 districts.

## Practice

Report the two fitted score differences for 22 → 20 students per teacher. Explain why they need not be the same.

**Explanation:** The simple model gives about +4.56 points. The adjusted model gives about +2.20 points while holding English-learner share fixed. The first comparison also reflects observed differences in language composition across districts; the second accounts for those differences within its stated model.

# Inspect variables before running the model

Check the data, then use a saved do-file to reproduce the estimates.

> Check observation counts, units, ranges and missing values.

> Outcome first; regressors afterward.

# Read a genuine Stata result

Read the coefficient, SE, test and interval in a real Stata table.

```
. quietly regress score str

. matrix list e(b), format(%10.6f)

e(b)[1,2]
           str       _cons
y1   -2.279808  698.932953
```

Actual Stata coefficient-only display; reproduce with coefficient_workshop.do.

> $\widehat{\beta}_1 = -2.279808$ points per student per teacher

> Fitted contrast: $(-2)(-2.279808) = 4.559616$ points

# Calculate the policy-sized fitted difference

lincom translates a slope into the particular comparison we want.

```
. display %10.6f -2*_b[str]
  4.559617
```

Actual Stata coefficient-only display; reproduce with coefficient_workshop.do.

$$\Delta X = 20 - 22 = -2$$

$$\Delta \widehat{Y} = \widehat{\beta}_1 \Delta X = 4.559616 \text{ points}$$

# Read the adjusted regression

Read both coefficients with their units and the stated comparison.

```
. quietly regress score str el

. matrix list e(b), format(%10.6f)

e(b)[1,3]
           str          el       _cons
y1   -1.101296   -0.649777  686.032250
```

Actual Stata coefficient-only display; reproduce with coefficient_workshop.do.

> Adjusted staffing slope: $-1.101296$

> At fixed language share: $(-2)(-1.101296) = 2.202592$ points

# Attempt the adjusted policy contrast

Keep the arithmetic and the interpretation together.

```
. display %10.6f -2*_b[str]
  2.202592
```

Actual Stata coefficient-only display; reproduce with coefficient_workshop.do.

$$\Delta \widehat{Y} = (-2)(-1.101296) = 2.202592 \text{ points}$$

### Practice

Calculate the adjusted fitted contrast for a ratio reduction from 22 to 20. Explain what is held fixed.

**Explanation:** About +2.20 score points, with English-learner share fixed in the fitted model.

# Would the result apply in a new setting?

Check credibility in the original setting and relevance to the new one.

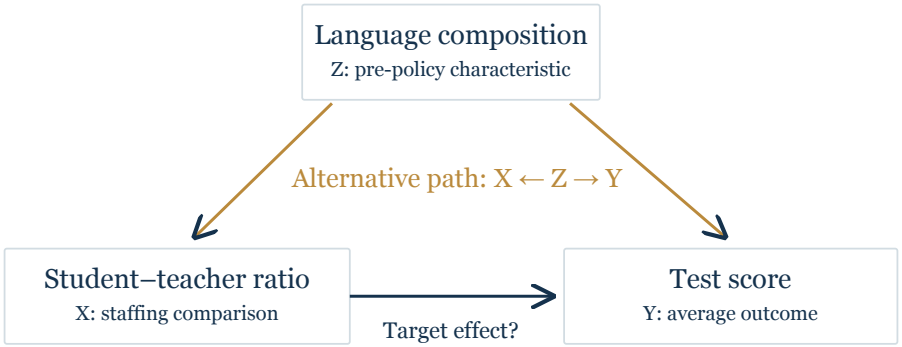
**Practice**

Propose a local staffing study. Name the schools or districts, policy change, learning measure, comparison and main concern.

**Explanation:** For example, study school-level learning gains after a funded reduction in the staffing ratio. Compare funded schools with schools whose lack of funding gives a credible estimate of what would otherwise have happened. If struggling schools are selected for funding, the comparison must address their different expected progress.

# Diagnose the claim

A test of the slope and a credible policy comparison answer different questions.




Proposed causal structure, not a relationship proved by the scatterplot.

Arrows show the proposed explanation. The scatterplot alone does not establish it.

## Practice

Rewrite: 'Significant slope proves that smaller classes cause higher scores.'

**Explanation:** Districts with lower student–teacher ratios have higher fitted scores in these data. A causal staffing effect would require a defensible comparison addressing factors such as family resources and staffing responses to student needs.

# What would convince you next?

Choose new data or a new comparison to address a specific concern.

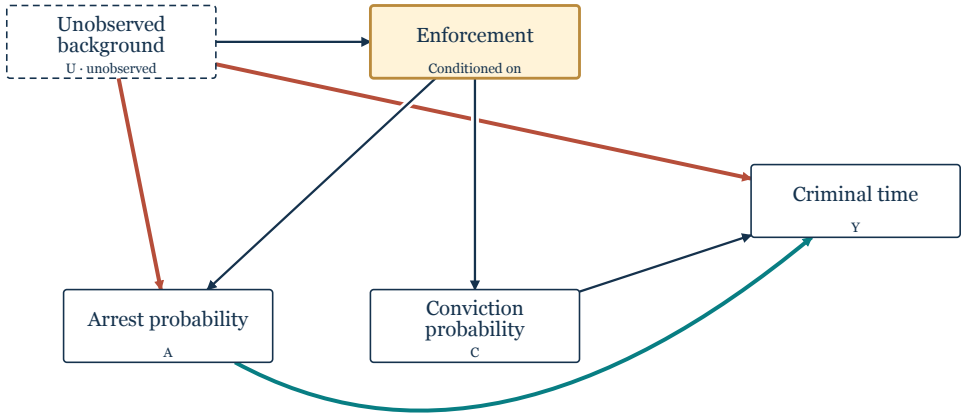
**Practice**

Compare adding more districts with allocating a staffing offer by chance. What can each improve?

**Explanation:** More districts can improve precision if the sampling conditions are suitable. Chance allocation can support a causal comparison by making the offer unrelated to other factors before the policy. We must still check whether the offer was implemented and whether the groups were affected by other differences.

# DAG practice: conviction and arrest

Trace the routes before choosing a control.




1 open directed route · 1 open alternative · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

**Practice**

In the law-and-order model, hold enforcement fixed. Which backdoor route from arrest probability to criminal time remains open? Explain why the same enforcement adjustment can be enough for the drawn conviction comparison but insufficient for arrest.

# DAG practice: conviction and arrest: explanation

Explain what conditioning changes in this stated graph.

> For arrest, the route $A \leftarrow U \rightarrow Y$ remains open because unobserved background affects both arrest probability and criminal time. Holding enforcement fixed does not block that route. For conviction, every drawn backdoor starts through enforcement, so conditioning on enforcement blocks those routes in this stated graph. The result is specific to the chosen explanatory variable and assumed arrows. If the real conviction process had another unmeasured parent affecting the outcome, that control set could also be insufficient.

# School staffing: reproduce and explain

Reproduce the evidence, then explain its economic meaning.

**Practice**

Add district income to the model already controlling for language share. Recalculate the fitted gain for a two-student reduction, with its robust SE. Keep your commands, log, output and interpretation.

# School staffing: reproduce and explain: explanation

A reproducible result needs both a calculation and an interpretation.

> Original two-student fitted gains: +4.56 and +2.20 points.

> With income: gain 0.137551, robust SE 0.580319.

> The smaller adjusted association does not establish a causal intervention effect.

# Exit check: explain one result carefully

Connect the observation, fitted difference, adjustment and policy question.

### Practice

Write the four-sentence report described here.

**Explanation:** One observation is a school district, and staffing is measured as students per teacher. A reduction of two students per teacher gives about +4.56 fitted score points in the simple model. The difference is about +2.20 points after holding English-learner share fixed. Other district differences or staffing responses to learning needs could still prevent either number from measuring the causal gain from hiring teachers.